

Mateo Rodríguez

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Experience

Technical Committee of FyT 2025 Congress 07/2025

I had the pleasure of helping organize the Fuerzas y Túnel Conference, a biennial event that brings together the Spanish and international scientific community to discuss the latest developments in scanning probe microscopy. Book of abstracts.

Research Internship: Surface Science 2024-2025

Working on Surface & Molecular Physics at SumoLab, a surface science research group which is part of the Nanomaterials and Nanotechnology Research Center (CINN).
Main experience: operating a Low Temperature Scanning Tunneling Microscope (LT STM) under conditions of low temperature ($T \approx 4$ K) and ultra-high vacuum ($P \sim 10^{-11}$ mbar).

Research Internship: Molecular & Computational Physics 2025

Project IFF-01 at the Fundamental Physics Institute (IFF), thanks to a JAE Intro ICU **competitive grant** funded by the Spanish National Research Council (CSIC).
Molecular Dynamics (MD) with Fortran and LAMMPS. Contribution to the open software MD package LAMMPS with a new intermolecular force field lj/pirani: https://doi.org/10.20350/digitalCSIC/17300.
In preparation:
"Permeation of hydrogen across graphdyine: molecular dynamics vs. quantum simulations and role of membrane motion"

Research Internship: Data Science 09-12/2024

Project SV-24-GIJON-1-15 with a **competitive grant** funded by IUTA. We created a Machine Learning (ML) model able to predict faults and performace of a marine diesel engine. Public talk's presentation here.

Applications of ML to the Study of Brain Metastases 04-06/2024

Python project developed inside the context of Saturdays AI Asturias. Github link Code link

Education

MSc in Theoretical Physics 2025-

Utrecht University

Bachelor of Science - Physics (240 ECTS) 2025

University of Oviedo
Bachelor Thesis: *Connecting theory and experiment: study of molecules on surfaces trough STM and simulations.*

VCQ Summer School: Interfacing Gravity and Quantum Physics 09/2024

25.5h course, organized by Vienna Center for Quantum Science and Technology. Certificate Link

Dutch Summer School of Theoretical Physics 08/2024

1.5 ECTS course on Theoretical Physics. Utrecht Summer School. Certificate Link

XII Gefenol Summer School on Statistical Physics of Complex Systems 07/2024

International 60h course on Complex Systems, applied Artificial Intelligence (AI) and Data Modeling. URJC Madrid. Certificate Link

Saturdays AI Asturias 02-06/2024

160h AI course with Python in which a project with social impact is developed. Certificate Link

Technical Skills

Python ★★★★☆

Fortran ★★★☆☆

Linux/Bash ★★★☆☆

C/C++ ★★☆☆☆

Molecular Dynamics

Computational Physics/ Modelling & Simulation

Data Science/Analysis

Languages

Spanish ★★★★★

Native

English ★★★★☆

Fluent (Level C2)

About Myself

I am passionate about science, I enjoy writing and I love football. Japanese art and culture fascinate me. I come from a family in which hard work and empathy are critical values. For more info about me, visit my website mateo.github.io.

Certifications

Cambridge Advanced 2025

Overall score of 203, Level C2. Certificate